

Swiggy Sales & Customer Analytics

End-to-End Data Analytics Workflow & Portfolio Project

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Tools	Excel / Python (Pandas), SQL, Power BI, GitHub
GitHub repository	https://github.com/aksharadeshmukh1311-crypto/swiggy-analytics



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1. Executive Summary

This project analyses Swiggy restaurant, city and customer data covering about 149K records across 821 cities (2017-2020). The data was cleaned, analysed with SQL and Python, and presented in a 5-page Power BI dashboard (Overview, User Performance, City Overview, Restaurant Analysis, Insights). The aim is to show where sales come from, who the key customers are, and what Swiggy should do next.

Headline findings

- **Top cities:** Tirupati leads all-time sales (43M), then Electronic City, Bangalore (29M), Baner, Pune (27M), Raipur (23M).
- **Customers:** Users aged 22-24 are the largest group (peak at about 2.1K users); students are 52.94% of the occupation mix on the 2020 female view.
- **Menu mix:** Veg leads on volume (156K vs 140K Non-Veg), but Non-Veg has the higher average price (231.81 vs 182.11).
- **Trend:** Quantity rose sharply from 2017 to a peak in 2018, then fell in 2019 and 2020.
- **Concentration:** The top 10% customers measure is 726.44M, about 75% of total sales of 963.83M.

2. Business Problem and Objectives

Business problem: Swiggy's management needs to understand which cities, restaurants, cuisines and customer segments drive sales, and why growth slowed after 2018, so marketing and expansion budget can be focused on the highest-value areas.

Objectives

- Identify top cities and restaurants by sales, users and ratings.
- Profile customers by age, gender, marital status and occupation.
- Measure year-over-year change in sales and quantity (2017-2020).
- Compare Veg, Non-Veg and Others on volume and average price.
- Measure how concentrated revenue is in the top 10% of customers.
- Turn findings into clear, actionable recommendations.

3. Dataset and Data Collection

Source: Swiggy restaurant / order / user dataset (add the dataset link or Kaggle URL here). The raw files were collected into one folder, merged on the common keys (city, restaurant, user) and organised into the structure shown in Appendix C.

Fields used in the analysis (as seen in the dashboard)

Group	Fields
Location	City (821 unique values)
Restaurant	Name, Cuisine, Rating, Rating Count, Veg/Non-Veg/Others
Transaction	Price, Sales (Amount), Quantity, Year, Orders
User	Age, Gender, Marital Status (Single/Married), Occupation (Student, Employee, Self Employed, House wife)

Dashboard headline metrics

Metric	Value	Metric	Value
Orders count	149K (148,577)	Total cities	821
User count	100K	Total sales	963,829,618
Rating count	61K	Restaurant count	148K
Avg price - Veg / Non-Veg / Others	182.11 / 231.81 / 19.86	Current_YR_Sale / Previous_YR_Sale	141M / 332M

4. Analytics Workflow and Tools

Stage	What was done	Tool
1. Business problem	Defined questions and KPIs	Documentation
2. Data collection	Gathered and organised raw files	Excel / CSV
3. Cleaning	Fixed types, duplicates, missing values, outliers	Python (Pandas) / Excel
4. EDA and analysis	Aggregations, YoY growth, segmentation, Pareto check	SQL, Python
5. Visualisation	5-page interactive dashboard with DAX measures	Power BI
6. Insights and report	Findings, recommendations, PDF report	PDF, Markdown
7. Portfolio	Repository, README, resume bullet	GitHub

5. Data Cleaning

Cleaning was done with Python/Pandas (full script in Appendix A). Each step protects the accuracy of the dashboard numbers.

Step	Action	Why it matters
Standardise text	Trim spaces, consistent casing for city, cuisine, gender, occupation	Prevents one city or cuisine appearing as several values
Fix data types	Convert price, sales, quantity, rating, age to numeric; year to integer	Enables correct sums and averages
Duplicates	Drop exact duplicate rows	Avoids double-counting orders and sales
Missing values	Drop rows without city or sales; fill missing rating with median	Keeps KPIs reliable

Step	Action	Why it matters
Invalid records	Remove negative sales or price; keep age within 10-90	Removes impossible values
Outliers	IQR rule on sales; flagged, not deleted blindly	Extreme values may be genuine large orders
Export	Save data/cleaned/swiggy_cleaned.csv	Single clean source for SQL and Power BI

6. Exploratory Data Analysis and Analysis (SQL and Python)

The analysis answers each objective with a query or a Pandas aggregation. Full SQL is in Appendix B and Python in Appendix A.

Question	Method	Output used in
Top cities by sales	GROUP BY city, SUM(sales), ORDER BY DESC	City Overview, Insights
Year-over-year change	SUM(sales) by year with LAG() window function	Overview (Quantity by Year)
Veg vs Non-Veg	COUNT, SUM(sales), AVG(price) by category	Overview, Restaurant Analysis
Age, gender, occupation	Segment counts and sales by group	User Performance
Top 10% customers	NTILE(10) on customer sales, share of total	Overview, Insights
Top cuisines and restaurants	COUNT DISTINCT restaurants by cuisine	Restaurant Analysis

7. Power BI Dashboard (5 pages)

The dashboard (file: swiggy_project.pbix) has a consistent Swiggy-orange theme, a side navigation bar with icons, and slicers for Amount/Quantity, Top N, Year, Gender and City.

Page 1 - Overview. KPI cards (orders, users, rating count), average price for Veg, Non-Veg and Others, a Top N city bar chart (Electronic City leads with 0.36M quantity) and a Quantity by Year line chart.



Figure 1: Overview page

Page 2 - User Performance. Year slicer (2017-2020) and Gender toggle. Shows Marital Status (Single 41M, Married 16M), Occupation donut (Student 32M / 52.94%, Employee 20M / 33.04%, Self Employed 5M / 8.48%) and Users by Age (peak around age 23).

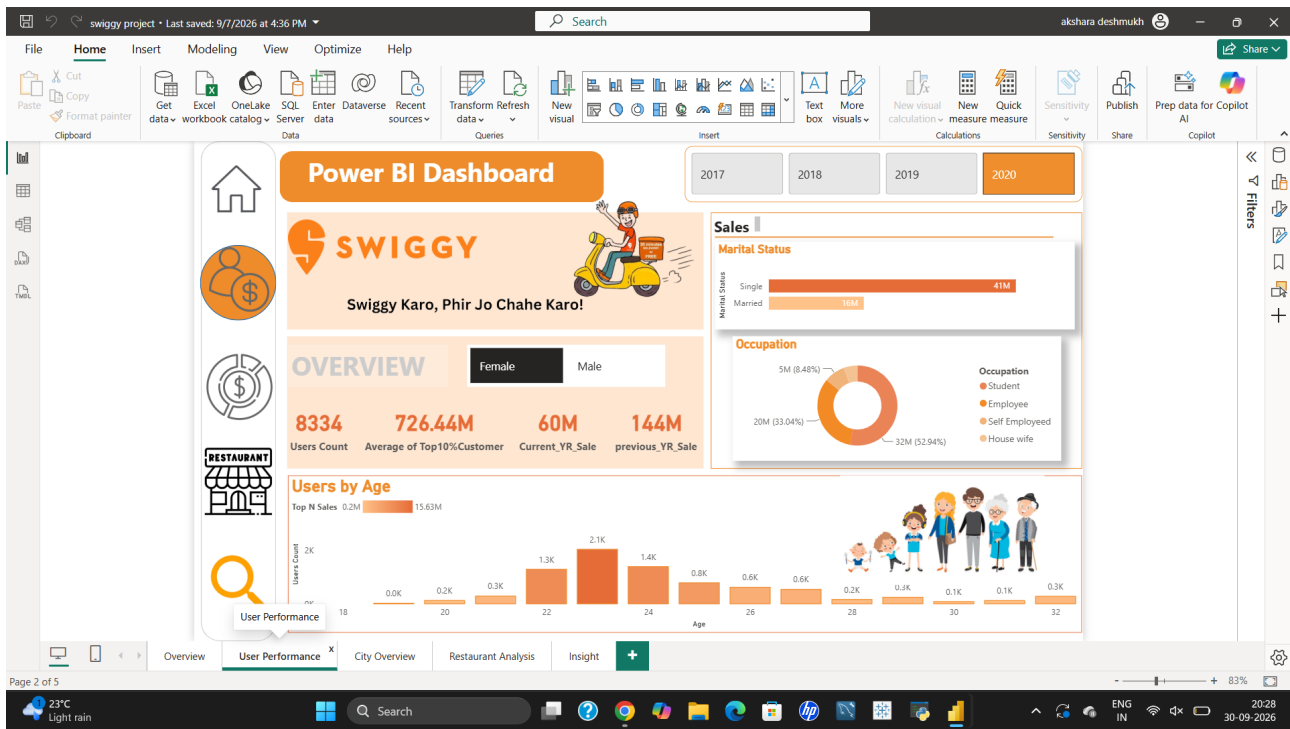


Figure 2: User Performance page (2020, Female filter)

Page 3 - City Overview. City slicer, a city table (sales, orders, current-year sales) and three bar charts for Sales, Users Count and Rating Count. Tirupati leads sales (43M); Bikaner leads users (1.7K) and ratings (850).



Figure 3: City Overview page

Page 4 - Restaurant Analysis. Veg / Non-Veg pie, Price by Cuisine, Restaurant Count by City, Top 5 Cuisines (North Indian 6.5K, Indian 6.4K, Chinese 5.1K) and top rated-count restaurants (Domino's 409, Pizza Hut 318, KFC 306).

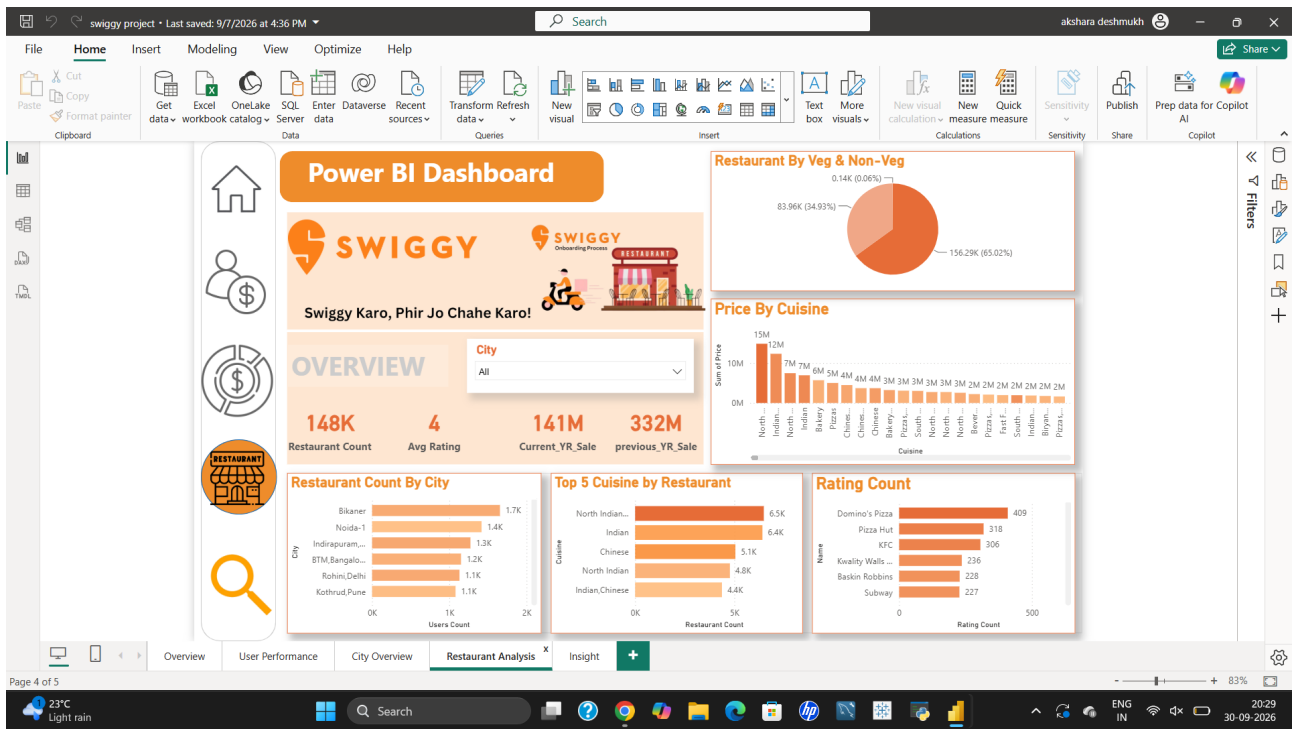


Figure 4: Restaurant Analysis page

Page 5 - Insights. Written findings and recommendations for Overview, User Performance and City Overview.

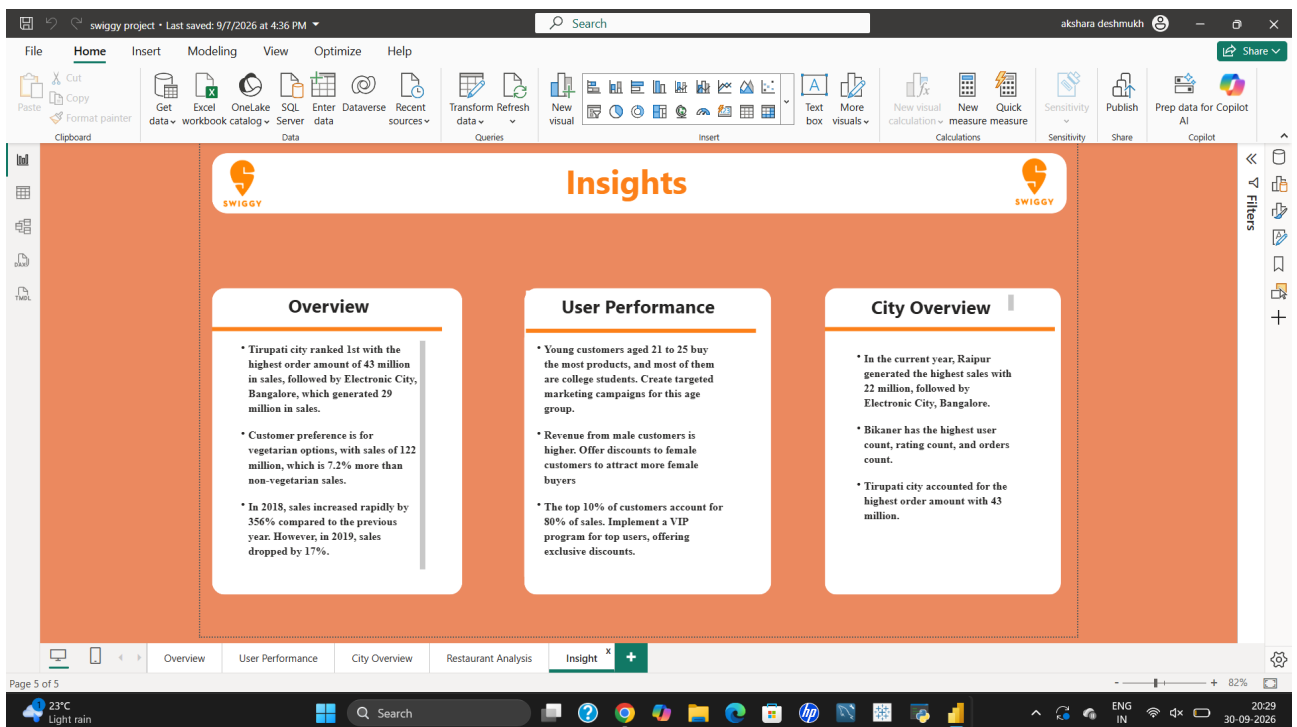


Figure 5: Insights page

8. Key Trends and Business Insights

#	Insight	Evidence (from dashboard)
1	Sales are concentrated in a few cities	Tirupati 43M, Electronic City 29M, Baner 27M, Raipur 23M out of 963.8M total across 821 cities
2	Bikaner is the engagement leader	Highest users (1.7K) and rating count (850), yet not top in sales - a monetisation gap
3	Young students drive demand	Users peak at ages 22-24; students are 52.94% of occupations in the 2020 female view
4	Singles spend far more than married users	Single 41M vs Married 16M (2020, female view)
5	Veg dominates volume, Non-Veg dominates price	156K vs 140K volume; average price 182.11 vs 231.81
6	Growth peaked in 2018 then declined	Quantity about 0.2M (2017) to about 1.0M (2018), then falling to about 0.35M (2020); Current_YR 141M vs Previous_YR 332M
7	Revenue depends on a small top segment	Top 10% customer measure 726.44M, about 75% of total sales
8	Chain restaurants lead in engagement	Domino's, Pizza Hut and KFC have the highest rating counts

9. Recommendations

- **Target the 21-25 segment:** run campaigns for college students (student offers, campus hubs, meal combos).
- **Launch a VIP programme:** reward the top 10% customers with exclusive discounts to protect the revenue they generate.
- **Grow female customers:** offer targeted discounts to raise female share (validate with both gender totals first).
- **Expand in proven cities:** strengthen partnerships in Tirupati, Electronic City (Bangalore), Baner (Pune) and Raipur.
- **Monetise Bikaner:** it has the most users and ratings but lower sales - test higher-value restaurants and promotions.
- **Promote Non-Veg premium items:** higher average price; bundle with Veg options, which drive volume.
- **Investigate the post-2018 decline:** review competition, discounts and data coverage before scaling spend.

10. Limitations and Next Steps

- The dataset covers 2017-2020 only; results may not reflect current market conditions.
- 2020 may be a partial year, which affects year-over-year comparisons.
- Customer segments (gender, occupation) are shown for a single filter view in the screenshots.
- Next: add a profit or margin field, run a cohort/retention analysis, and build a simple forecast of sales.

Appendix A: Python script (clean_and_eda.py)

```
"""Swiggy project - data cleaning and EDA.
Adjust the COLS mapping to match the column names in your raw file."""
import pandas as pd

RAW = "data/raw/swiggy_raw.csv"
OUT = "data/cleaned/swiggy_cleaned.csv"

# Left: name used below | Right: name in YOUR raw file (edit these)
COLS = {
    "city": "City", "restaurant": "Name", "cuisine": "Cuisine",
    "rating": "Rating", "rating_count": "Rating_Count", "price": "Price",
    "diet": "Veg_or_Non_Veg", "sales": "Sales", "quantity": "Quantity",
    "year": "Year", "age": "Age", "gender": "Gender",
    "marital": "Marital_Status", "occupation": "Occupation",
    "user_id": "User_ID",
}

df = pd.read_csv(RAW)
df = df.rename(columns={v: k for k, v in COLS.items() if v in df.columns})
print("Raw shape:", df.shape)

# 1. Standardise text columns
for c in ["city", "restaurant", "cuisine", "diet", "gender",
          "marital", "occupation"]:
    if c in df.columns:
        df[c] = df[c].astype(str).str.strip().str.title()

# 2. Fix data types
for c in ["rating", "rating_count", "price", "sales", "quantity", "age"]:
    if c in df.columns:
        df[c] = pd.to_numeric(df[c], errors="coerce")

# 3. Missing values and duplicates
print("Missing values:\n", df.isna().sum())
df = df.drop_duplicates()
df = df.dropna(subset=["city", "sales"])
if "rating" in df.columns:
    df["rating"] = df["rating"].fillna(df["rating"].median())

# 4. Remove invalid records
df = df[(df["sales"] >= 0) & (df["price"] >= 0)]
df = df[df["age"].between(10, 90) | df["age"].isna()]

# 5. Outlier check (IQR) on sales - flag, do not delete blindly
q1, q3 = df["sales"].quantile([0.25, 0.75])
iqr = q3 - q1
df["sales_outlier"] = ~df["sales"].between(q1 - 1.5 * iqr, q3 + 1.5 * iqr)
print("Sales outliers flagged:", int(df["sales_outlier"].sum()))

df.to_csv(OUT, index=False)
print("Cleaned shape:", df.shape)

# ----- EDA -----
print("\nTop 10 cities by sales")
print(df.groupby("city")["sales"].sum().nlargest(10))

print("\nQuantity by year")
qty = df.groupby("year")["quantity"].sum()
print(qty)
print("YoY % change:\n", (qty.pct_change() * 100).round(1))

print("\nAvg price and count by diet type")
print(df.groupby("diet").agg(count=("price", "size"),
                             avg_price=("price", "mean")).round(2))

print("\nUsers by age (top 10)")
print(df.groupby("age")["user_id"].nunique().nlargest(10))

print("\nSales by occupation and gender")
print(df.groupby("occupation")["sales"].sum().sort_values(ascending=False))
print(df.groupby("gender")["sales"].sum())

# Top 10% customers share of sales (Pareto check)
cust = df.groupby("user_id")["sales"].sum().sort_values(ascending=False)
top_n = max(1, int(len(cust) * 0.10))
share = cust.head(top_n).sum() / cust.sum() * 100
print(f"\nTop 10% customers contribute {share:.1f}% of total sales")
```

Appendix B: SQL queries (analysis.sql)

```
-- Swiggy project - analysis queries (MySQL / PostgreSQL compatible).
-- Table: swiggy_data(city, restaurant_name, cuisine, rating, rating_count,
-- price, veg_nonveg, sales, quantity, order_year, user_id, age, gender,
-- marital_status, occupation) -- rename to match your table

-- 1. Top 10 cities by sales
SELECT city, SUM(sales) AS total_sales, COUNT(*) AS orders
FROM swiggy_data GROUP BY city
ORDER BY total_sales DESC LIMIT 10;

-- 2. Year-over-year sales growth
WITH yearly AS (
    SELECT order_year, SUM(sales) AS total_sales
    FROM swiggy_data GROUP BY order_year)
SELECT order_year, total_sales,
    ROUND(100.0 * (total_sales - LAG(total_sales) OVER (ORDER BY order_year))
    / LAG(total_sales) OVER (ORDER BY order_year), 1) AS yoy_pct
FROM yearly ORDER BY order_year;

-- 3. Veg vs Non-Veg: volume, sales, average price
SELECT veg_nonveg, COUNT(*) AS records, SUM(sales) AS total_sales,
    ROUND(AVG(price), 2) AS avg_price
FROM swiggy_data GROUP BY veg_nonveg;

-- 4. Users by age group
SELECT CASE WHEN age < 18 THEN 'Under 18'
    WHEN age BETWEEN 18 AND 20 THEN '18-20'
    WHEN age BETWEEN 21 AND 25 THEN '21-25'
    WHEN age BETWEEN 26 AND 30 THEN '26-30'
    ELSE '31+' END AS age_group,
    COUNT(DISTINCT user_id) AS users, SUM(sales) AS total_sales
FROM swiggy_data GROUP BY 1 ORDER BY users DESC;

-- 5. Sales by gender and occupation
SELECT gender, occupation, SUM(sales) AS total_sales
FROM swiggy_data GROUP BY gender, occupation
ORDER BY total_sales DESC;

-- 6. Top 10% customers: share of total sales
WITH cust AS (
    SELECT user_id, SUM(sales) AS s,
        NTILE(10) OVER (ORDER BY SUM(sales) DESC) AS decile
    FROM swiggy_data GROUP BY user_id)
SELECT ROUND(100.0 * SUM(CASE WHEN decile = 1 THEN s END) / SUM(s), 1)
    AS top10pct_share
FROM cust;

-- 7. Top 5 cuisines by restaurant count
SELECT cuisine, COUNT(DISTINCT restaurant_name) AS restaurants
FROM swiggy_data GROUP BY cuisine ORDER BY restaurants DESC LIMIT 5;

-- 8. Cities with the highest user and rating counts
SELECT city, COUNT(DISTINCT user_id) AS users, SUM(rating_count) AS ratings
FROM swiggy_data GROUP BY city ORDER BY users DESC LIMIT 10;
```

Appendix C: GitHub README (README.md)

Swiggy Sales & Customer Analytics - End-to-End Data Analytics Project

An end-to-end analytics project on Swiggy restaurant, city and customer data: data collection, cleaning, SQL/Python analysis, a 5-page Power BI dashboard, business insights and recommendations.

1. Business Problem

Swiggy's business team needs to know where sales come from, who the most valuable customers are, and which cities and cuisines to prioritise.

The project answers:

1. Which cities and restaurants drive the most sales, users and ratings?
2. Who are the core customers (age, gender, occupation, marital status)?
3. How have sales and quantity changed year over year (2017-2020)?
4. How do Veg, Non-Veg and Other categories compare on volume and price?
5. How concentrated is revenue in the top 10% of customers?

2. Dataset

- Source: Swiggy restaurant / order / user dataset (ADD LINK HERE)
- Coverage: about 149K records, 821 cities, years 2017-2020
- Key fields: City, Restaurant Name, Cuisine, Rating, Rating Count, Price, Veg/Non-Veg, Sales (Amount), Quantity, Year, User Age, Gender, Marital Status, Occupation

3. Tools Used

Stage	Tool
Cleaning	Python (Pandas) / Excel
Analysis	SQL, Python
Visualisation	Power BI (DAX measures)
Documentation	Markdown, PDF, GitHub

4. Workflow

Raw data -> Cleaning -> EDA -> SQL/Python analysis -> Power BI -> Insights
-> Report -> GitHub.

5. Dashboard Pages

1. Overview - orders, users, avg price by Veg/Non-Veg/Others, top 10 cities, quantity by year
2. User Performance - age, gender, marital status, occupation, year slicer
3. City Overview - sales, users and ratings by city, city slicer
4. Restaurant Analysis - veg vs non-veg mix, cuisines, top restaurants
5. Insights - written findings and recommendations

6. Key Insights

- Tirupati leads sales (43M), followed by Electronic City, Bangalore (29M).
- Customers aged 21-25 (mostly students) are the largest user group.
- Veg options lead on sales; Non-Veg has the higher average price.
- Sales grew sharply in 2018 and declined in 2019 and 2020.
- Revenue is heavily concentrated in the top 10% of customers.

7. Recommendations

1. Run targeted campaigns for the 21-25 student segment.
2. Launch a VIP programme for top-spending customers.
3. Offer discounts to attract more female customers.
4. Expand partnerships in high-sales cities (Tirupati, Raipur, Bangalore).
5. Investigate the post-2018 decline before scaling marketing spend.

8. Repository Structure

```
swiggy-analytics/  
|-- data/  
|   |-- raw/           original dataset  
|   |-- cleaned/       cleaned dataset (CSV)  
|-- notebooks/         EDA notebook  
|-- sql/               analysis queries  
|-- python/            clean_and_eda.py  
|-- powerbi/           swiggy_project.pbix  
|-- screenshots/       dashboard pages (PNG)  
|-- reports/           final PDF report  
|-- README.md  
...
```

9. How to Run

1. Place the raw file in data/raw/.
2. Run: python python/clean_and_eda.py
3. Load the cleaned CSV into your SQL database and run sql/analysis.sql.
4. Open powerbi/swiggy_project.pbix and refresh the data source.

10. Author

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GitHub: <https://github.com/aksharadeshmukh1311-crypto/swiggy-analytics>